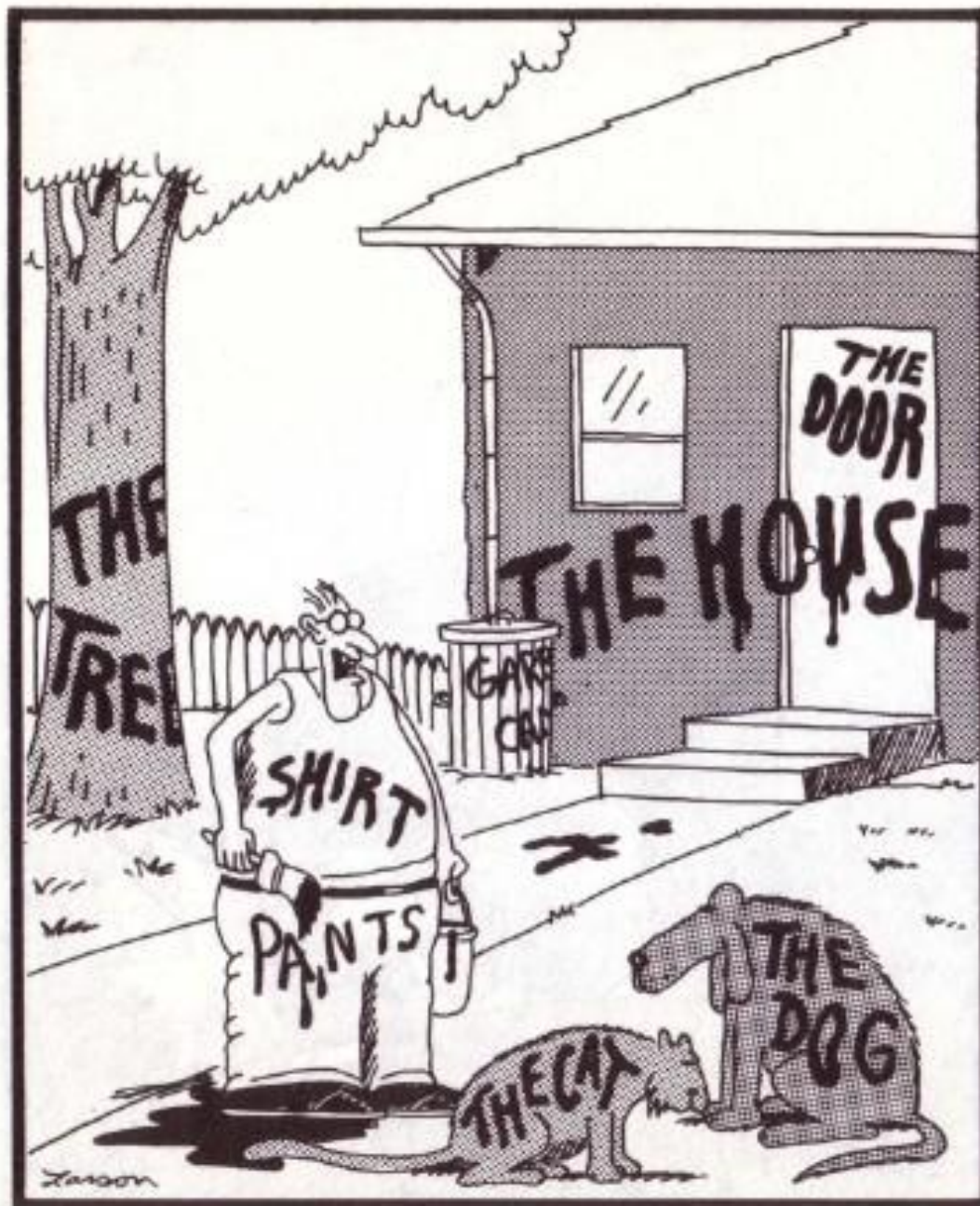




"Now! ... *That* should clear up
a few things around here!"

- "Dobrý den pane. Kolik je hodin?"
- "To je přesně 08:32."

- "Good morning sir. What time is it ? "
- "It's 8:32 am."

- "Good morning sir. What time is it ? "
- "It's 8:32 am."



- "I have a meeting at 8:30 am. I'm in late. I need to run."



- "Good morning sir. What time is it ? "
- "It's 8:32 am."

```
transport= "foot",  
transport_speed= "walking",  
present_time = 8:32,  
meeting_time = 8:30,  
if present_time > meeting_time then situation = "late",  
if situation = "late" and transport = "foot"  
then transport_speed = "run"  
and words="I have a meeting at "+meeting_time+". I'm in  
"+situation+". I need to "+transport_speed+"."
```

- "I have a meeting at 8:30 am. I'm in late. I need to run."



What are the films in which Harrison Ford plays the role of a scientist ?

What will be your approach ?

- We can try with a Google query like :
"Harrison Ford plays scientist"

- And if we had an ideal Data base

movie_title	actor	character
Star Wars:Episode VII	Harrison Ford	Han Solo
Indiana Jones and the Last Crusade	Harrison Ford	Indiana Jones
Sabrina	Harrison Ford	Linus Larrabee

character	job
Han Solo	smuggler
Indiana Jones	archeologist
Linus Larrabee	lawyer

```
SELECT  c.movie_title, c.character
FROM CAST c, MOVIE_CHARACTER mc
WHERE  c.actor="Harrison Ford"
      AND c.character=mc.character
      AND (mc.job="scientist" OR mc.job="physicist" OR
mc.job="biologist" mc.job="professor" OR
mc.job="physician" OR ... )
```

```
<actor>Harrison Ford</title>
<filmography>
  <movie>Star Wars : Episode VII</movie>
  <movie>Indiana Jones and the Last Crusade</movie>
  <movie>Sabrina</movie>    ...
</filmography>
```

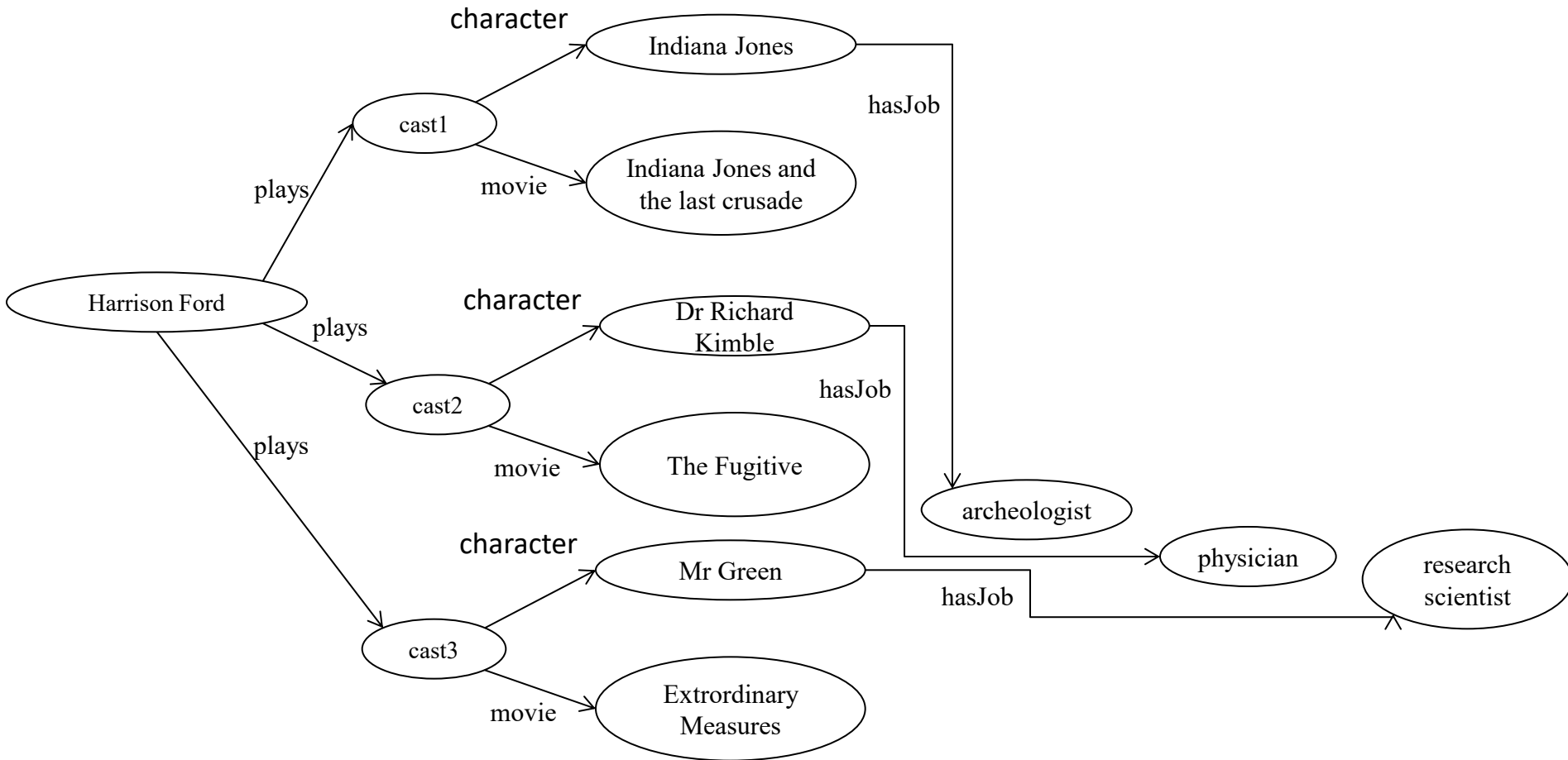
```
<title>Indiana Jones and the Last Crusade</title>
<full cast>
  <cast>Harrison Ford    ...    Indiana Jones</cast>
  <cast>Sean Connery     ...    Professor Henry Jones</cast>
  <cast>Denholm Elliott  ...    Marcus Brody</cast>    ...
</full cast>
```

```
<character>Indiana Jones</character>
<biography>
Dr. Henry "Indiana" Jones, Jr. (born July 1, 1899), also known as
Indy, is a fictional professor, archaeologist, and adventurer - the
main protagonist of the 1981 adventure film Raiders of the Lost Ark
(later retitled Indiana Jones and the Raiders of the Lost Ark) ...
</biography>
```

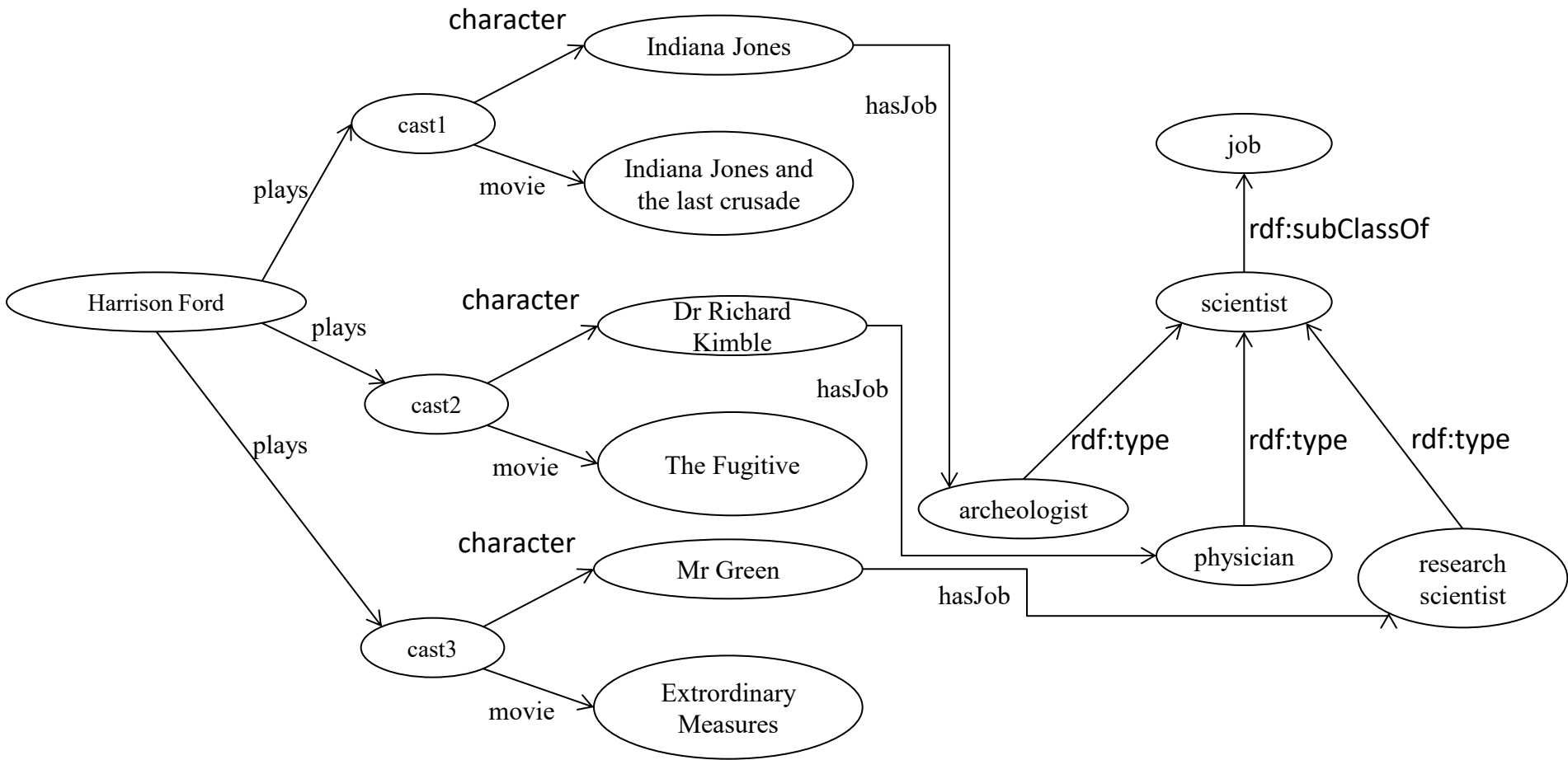
One has to be able to browse the Web, to read, and to know
that an archeologist (or a Dr. or a professor) is a scientist

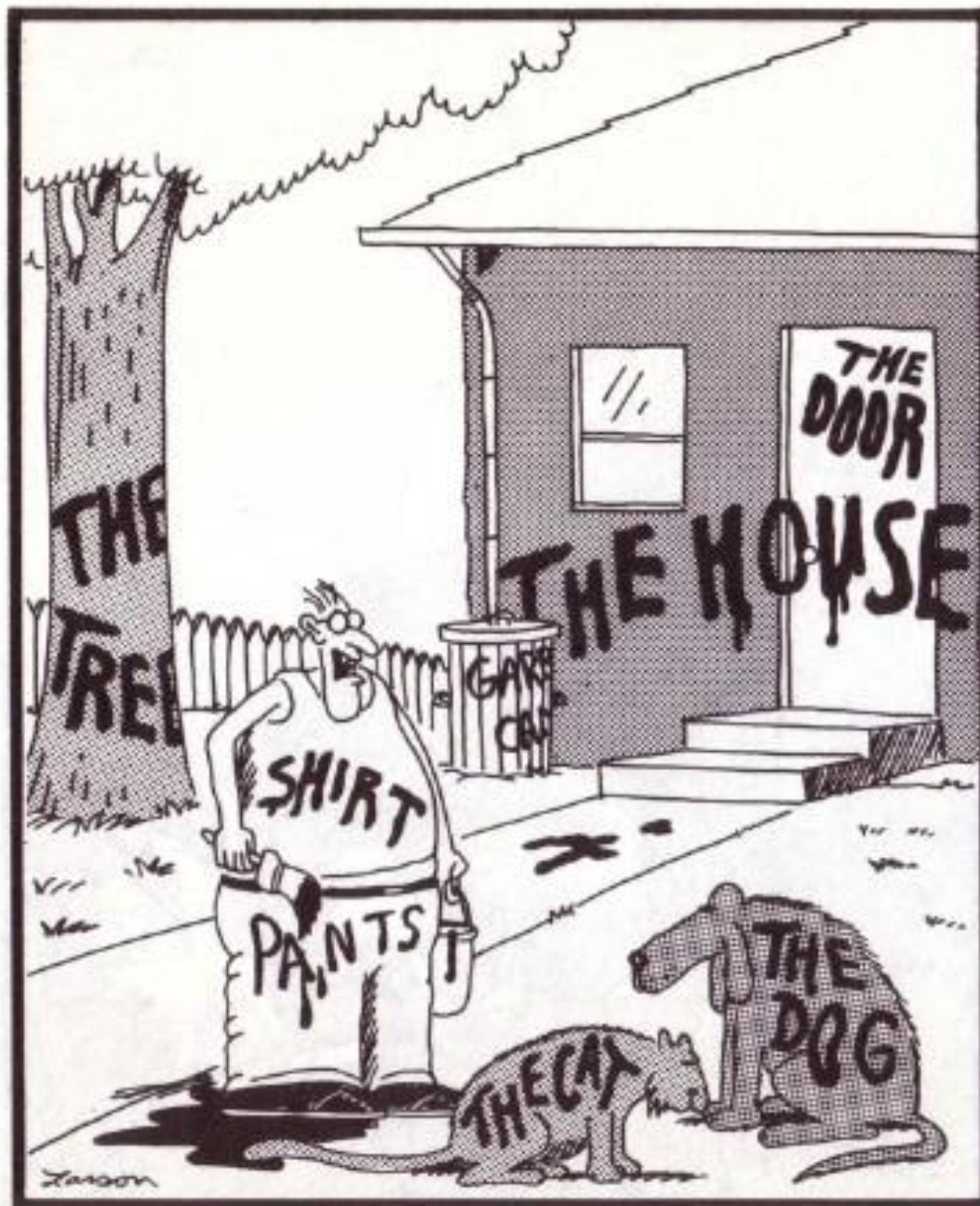
Quite good ... but it doesn't let answer our query

The thing proposed by the Semantic web is:



Ce que propose le web sémantique :





"Now! ... *That* should clear up
a few things around here!"

Representing things is good ...

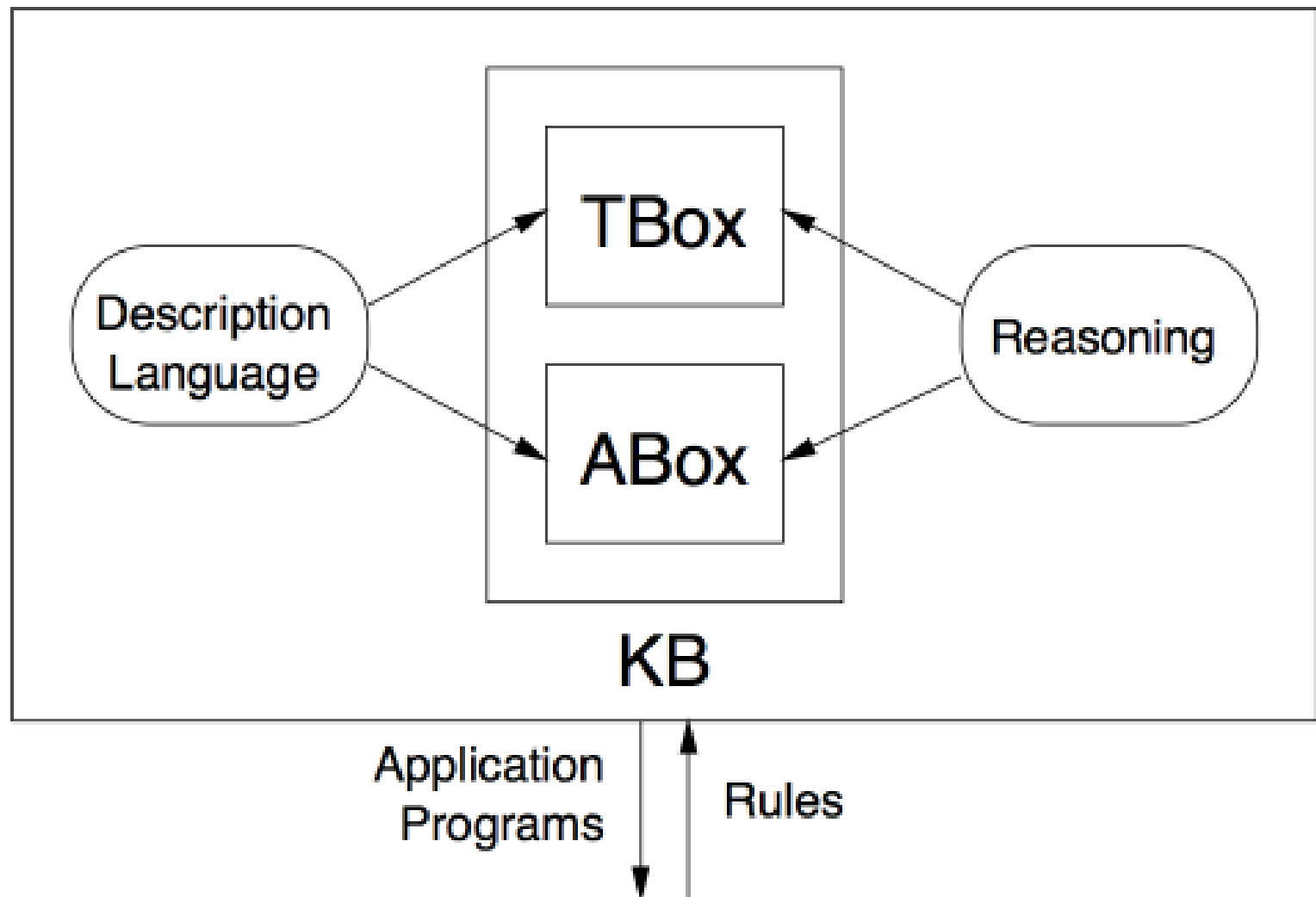
Isn't it ?

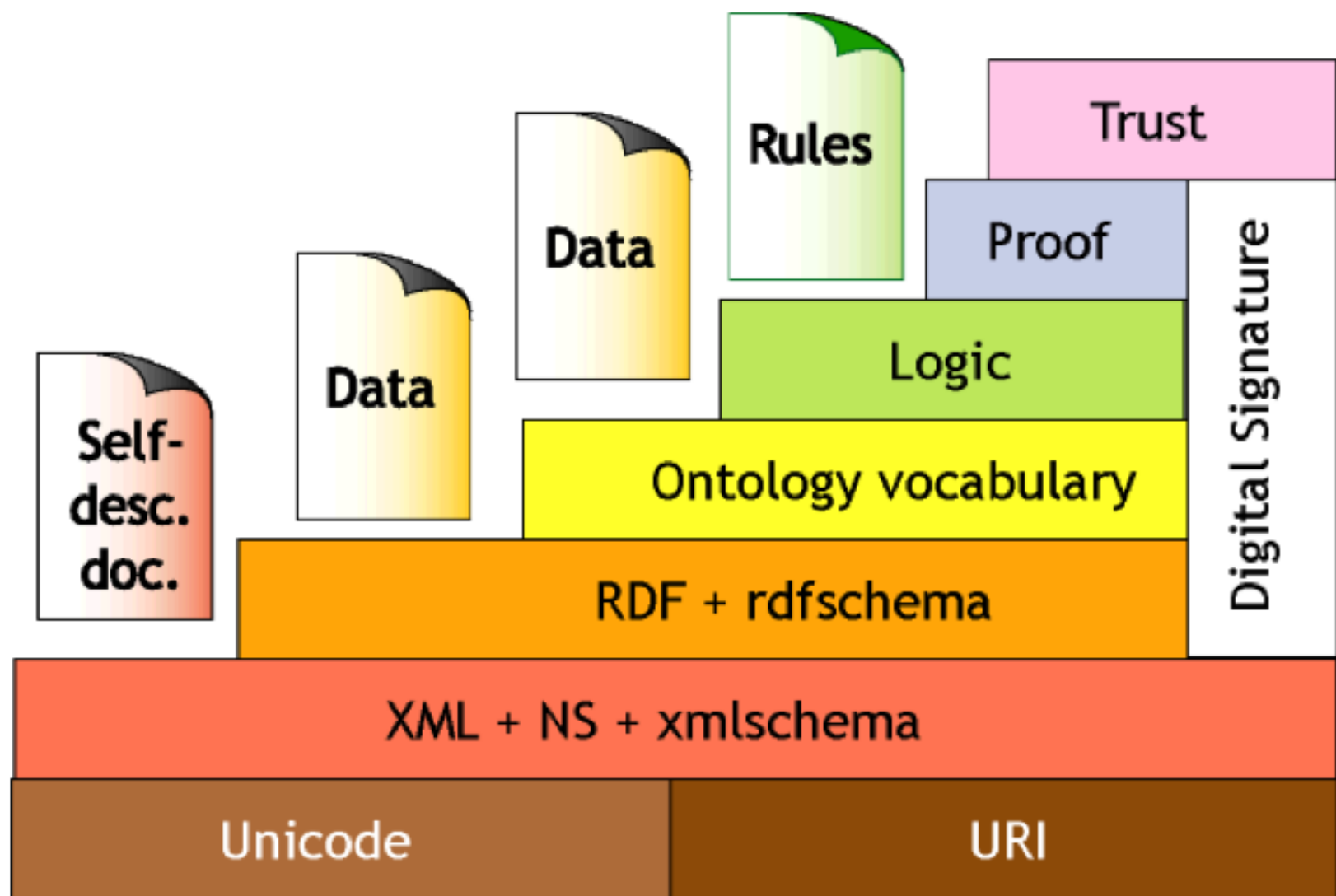
... Tell me some form of representation that you know ...

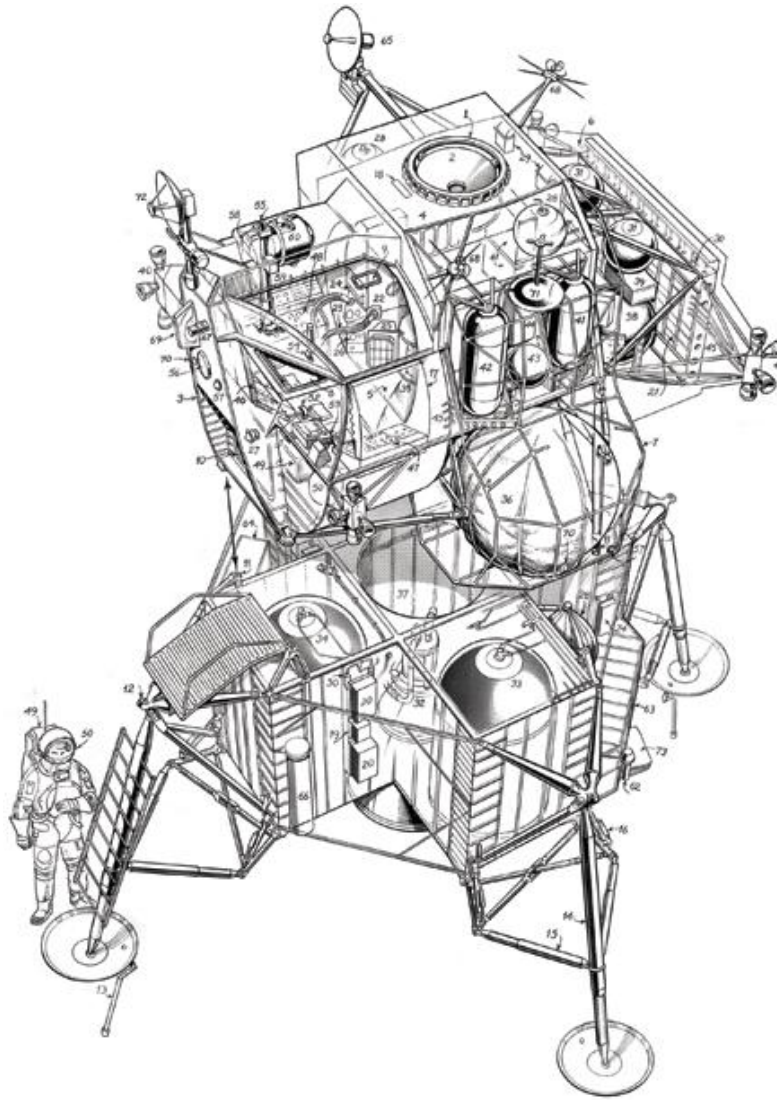
- Representing things is good ...
- Doing it in a standard way is better ...
- Doing it following a precise formalism which is associated to some reasoning mechanisms is even better.

- We talk then about ...

Knowledge base systems







module lunaire =
15 million de pièces